ZHAOMIN WU

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RESEARCH PROFILE

I am an NRF Postdoctoral Fellow in Computer Science at the National University of Singapore. My research develops trustworthy AI systems, with a focus on federated learning, machine unlearning, multi-agent systems, and AI for Psychology. My work has appeared in ICLR, NeurIPS, SIGMOD, ACL, KDD, and related venues, including two ICLR 2026 Oral papers and a SIGMOD 2023 Honorable Mention for Best Artifact. As of May 2026, my work has received 2,710 citations on Google Scholar.

EMPLOYMENT

National University of Singapore (NUS), Singapore 05/2024 – Present

- *NRF Postdoctoral Fellow* (04/2026 – Present), **Principal Investigator (PI)** of an NRF-funded postdoctoral project at Department of Computer Science.
- *Research Fellow* (05/2024 – 03/2026), Institute of Data Science / Department of Computer Science; Project PIs: Prof. Ng See Kiong, Prof. Bingsheng He.

EDUCATION

National University of Singapore (NUS), Singapore 07/2019 – 05/2024
Ph.D. in Computer Science, *Honorable Mention for Best Ph.D. Thesis* Advisor: Prof. Bingsheng He

Huazhong University of Science and Technology (HUST), China 09/2015 – 06/2019
B.Eng. in Computer Science and Technology, ACM class GPA: 3.92/4.0, Rank: 10/281

GRANTS & FELLOWSHIPS

Singapore NRF Postdoctoral Award (2026), National Research Foundation, Singapore
Principal Investigator with four-year salary support and an ***independent SGD 250,000 research grant***; an exceptionally competitive national-level fellowship.

Google Cloud Research Credit Award (2026), Google
USD 5,000 cloud credits for research computing.

SELECTED HONORS & RECOGNITION

Honorable Mention for Best Ph.D. Thesis Award (2024), NUS School of Computing (5/∼150).

Best Research Staff Award (2025), NUS Institute of Data Science (2/∼50).

Dean's Graduate Research Excellence Award (2023), NUS School of Computing (10/∼150).

SELECTED PUBLICATIONS

Z. Wu, H. Zhao, Z. Wang, J. Guo, Q. Wang, B. He. **LLM DNA: Tracing Model Evolution via Functional Representations**. (**ICLR'26**). ***Oral presentation, top 1% of submissions.*** *Covered by XinZhiYuan, Sina Finance, Sohu, BAAI Hub, and NICE Academic; NICE Academic video recorded 1,000+ views.*

Z. Wu, M. Du, S.-K. Ng, B. He. **Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts**. (**ICLR'26**). ***Oral presentation, top 1% of submissions.*** *Covered by JiQiZhiXin and AITNT; AITNT post recorded 6,500+ clicks.*

Z. Wu, J. Zhu, Q. Li, B. He. **DeltaBoost: Gradient Boosting Decision Trees with Efficient Machine Unlearning**. (**SIGMOD'23**). ***Honorable Mention for Best Artifact, one of 3 selected from ∼600 SIGMOD submissions.*** *The first GBDT-like model to support efficient machine unlearning.*

Z. Wu, J. Hou, B. He. **VertiBench: Advancing Feature Distribution Diversity in Vertical Federated Learning Benchmarks**. (**ICLR'24**). *The first benchmark to systematically evaluate vertical federated learning under varying feature-distribution settings.*

Z. Wu, Q. Li, B. He. **A Coupled Design of Exploiting Record Similarity for Practical Vertical Federated Learning**. (**NeurIPS'22**). *Enables practical vertical federated learning by exploiting inter-party record similarity instead of exact identifier matching.*

RESEARCH IMPACT platform metrics recorded in May 2026

Open-Source Systems and Benchmarks

- **FedTree** (MLSys'23): an open-source federated learning system for tree-based models; it supports reproducible and privacy-preserving GBDT training across parties. 152 GitHub stars and 41 forks.
- **VertiBench** (ICLR'24): a public benchmark for systematically evaluating vertical federated learning under varying feature-diversity settings; it standardizes evaluation beyond single synthetic data splits and is released as an installable PyPI package.
- **WikiDBGraph** (ICDE'26): a large-scale database graph of Wikidata for collaborative learning; it releases 100,000 database nodes and 17 million edges for database-silo learning research.
- **LLM DNA** (ICLR'26 Oral): an open-source toolkit for extracting functional DNA signatures and tracing LLM evolution from model behavior, with a public project website and PyPI package.
- **OARF** (ACM TIST'22): a benchmark suite for characterizing federated learning systems; it provides reusable workloads for system-level FL comparison.

Publicity and Public Outreach

LLM DNA: Tracing Model Evolution via Functional Representations (ICLR'26 Oral). Media article by XinZhiYuan (新智元): "Testing DNA for Large Models: Hidden Lineage from Fine-tuning and Distillation Can No Longer Hide" (给大模型验DNA，微调、蒸馏的隐藏血缘藏不住了); reposted by Sina Finance (新浪财经), Sohu (搜狐), BAAI Hub (智源社区). Public outreach hosted by NICE Academic (NICE学术) on Bilibili: "Doing Paternity Testing for Large Models: LLM DNA and Model Evolution Graph" (给大模型做亲子鉴定：LLM 的DNA 与模型演化图谱).

Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts (ICLR'26 Oral). Media article by JiQiZhiXin (机器之心): "ICLR 2026 Oral: Large Models Can Still 'Lie' Without Prompt Induction" (ICLR 2026 Oral丨没人诱导，大模型也会骗人); reposted by AITNT.

Mining Intrinsic Rewards from LLM Hidden States for Efficient Best-of-N Sampling (KDD'26). Media article by XinZhiYuan (新智元): "Reward Models Are Changing: 0.005% Parameters, Faster Inference, Stronger Performance" (奖励模型变天！0.005%参数量推理速度翻倍，性能还更强); reposted by Sina Finance (新浪财经), Sohu (搜狐), BAAI Hub (智源社区), Tencent Cloud Developer Community (腾讯云开发者社区).

INVITED TALKS & KEYNOTES

Invited talk at **Hong Kong Baptist University**. Hosted by **Prof. Amelie Chi Zhou**. "Managing Data and Model Silos for Real-World AI Systems." (Jul. 2026)

Invited talk at **Université de Montréal (UdeM) and Mila**. Hosted by **Prof. Bang Liu**. "When Data and Models Stay Hidden: Toward Trustworthy AI Collaboration." (May 2026)

Keynote at **AAAI 2026 FLCA Workshop**. "Bridging Data Silos with Practical Federated Learning." (Jan. 2026)

Invited talk at **DASFAA 2025 Trust Day**, on behalf of Prof. Bingsheng He. "Towards Practical Vertical Federated Learning Systems." (May 2025)

Invited talk at **NUS Open House**. "Bridging Private Data Silo with Machine Learning." (Feb. 2025)

TEACHING & MENTORSHIP

Teaching Assistant, NUS

Introduction to Operating Systems (CS2106, Fall 2022); Software Testing (CS4218, Spring 2022); Neural Networks and Deep Learning (CS5242, Fall 2021); Programming Methodology II (CS2030, Spring 2021); Big Data Systems for Data Science (CS4225, Fall 2020); Data Structures and Algorithms (CS2040S, Spring 2020); Programming Methodology (CS1010E, Fall 2019).

Mentored Students

Undergraduate mentees (3 students from NUS and SJTU) 2024 – 2026
Representative: Jizhou Guo (SJTU) — Ph.D. offers from NUS and UIUC

Master's mentees (5 students from NUS) 2022 – 2026
Representative: Junyi Hou (NUS) — Ph.D. student, NUS

Visiting Ph.D. mentees (2 students from SJTU and ZJU) 2024 – 2025
Representative: Zhen Qin (ZJU) — Assistant Professor, ZJU

FULL PUBLICATION LIST

*Equal contribution, †Corresponding author

2026

[ICLR'26] Zhaomin Wu, Haodong Zhao, Ziyang Wang, Jizhou Guo, Qian Wang, Bingsheng He. LLM DNA: Tracing Model Evolution via Functional Representations. In The Fourteenth International Conference on Learning Representations (ICLR), 2026. ***(Oral, 1%)***

[ICLR'26] Zhaomin Wu, Mingzhe Du, See-Kiong Ng, Bingsheng He. Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts. In The Fourteenth International Conference on Learning Representations (ICLR), 2026. ***(Oral, 1%)***

[ICDE'26] Zhaomin Wu*, Ziyang Wang*, Bingsheng He. WikiDBGraph: Large-Scale Database Graph of Wikidata for Collaborative Learning. In Proceedings of the IEEE International Conference on Data Engineering (ICDE), 2026.

[KDD'26] Jizhou Guo, Zhaomin Wu†, Hanchen Yang, Philip S. Yu. Mining Intrinsic Rewards from LLM Hidden States for Efficient Best-of-N Sampling. In Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026.

[WWW'26] Yicheng Zhang, Zhen Qin, Zhaomin Wu, Shuiguang Deng†. Personalized Federated Fine-Tuning for LLMs via Data-Driven Heterogeneous Model Architectures. In Proceedings of the ACM Web Conference (WWW), 2026. ***(Oral, 9%)***

2025

[EMNLP'25] Zhaomin Wu*, Jizhou Guo*, Junyi Hou, Bingsheng He, Lixin Fan, Qiang Yang. Model-based Large Language Model Customization as Service. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[ACL'25] Zhen Qin, Zhaomin Wu†, Bingsheng He, Shuiguang Deng†. Federated Data-Efficient Instruction Tuning for Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2025.

2024

[NeurIPS'24] Zhaomin Wu, Junyi Hou, Yiqun Diao, Bingsheng He. Federated Transformer: Multi-Party Vertical Federated Learning on Practical Fuzzily Linked Data. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

[ICLR'24] Zhaomin Wu, Junyi Hou, Bingsheng He. VertiBench: Advancing Feature Distribution Diversity in Vertical Federated Learning Benchmarks. In The Twelfth International Conference on Learning Representations (ICLR), 2024.

2023

[SIGMOD'23] Zhaomin Wu, Junhui Zhu, Qinbin Li, Bingsheng He. DeltaBoost: Gradient Boosting Decision Trees with Efficient Machine Unlearning. In Proceedings of the ACM on Management of Data (SIGMOD), 2023. ***(Best Artifact Honorable Mention, 3 out of ∼600)***

[MLSys'23] Qinbin Li, Zhaomin Wu, Yanzheng Cai, Yuxuan Han, Ching Man Yung, Tianyuan Fu, Bingsheng He. FedTree: A Federated Learning System for Trees. In Proceedings of Machine Learning and Systems (MLSys), 2023.

2022

[TKDE'22] Qinbin Li, Zeyi Wen, Zhaomin Wu, Sixu Hu, Naibo Wang, Yuan Li, Xu Liu, Bingsheng He. A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy and Protection. IEEE Transactions on Knowledge and Data Engineering (TKDE), 2022.

[NeurIPS'22] Zhaomin Wu, Qinbin Li, Bingsheng He. A Coupled Design of Exploiting Record Similarity for Practical Vertical Federated Learning. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

[TBD'22] Zhaomin Wu, Qinbin Li, Bingsheng He. Practical Vertical Federated Learning with Unsupervised Representation Learning. IEEE Transactions on Big Data, 2022.

[TIST'22] Sixu Hu, Yuan Li, Xu Liu, Qinbin Li, Zhaomin Wu, Bingsheng He. The OARF Benchmark Suite: Characterization and Implications for Federated Learning Systems. ACM Transactions on Intelligent Systems and Technology (TIST), 2022.

2020

[AAAI'20] Qinbin Li, Zhaomin Wu, Zeyi Wen, Bingsheng He. Privacy-Preserving Gradient Boosting Decision Trees. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2020.

PREPRINTS

*Equal contribution, †Corresponding author

2026

[arXiv'26] Haodong Zhao, Jinming Hu, Zhaomin Wu†, Zongru Wu, Wei Du, Junyi Hou, Caibei Zhao, Zhuosheng Zhang, Bingsheng He, Gongshen Liu†. ProtegoFed: Backdoor-Free Federated Instruction Tuning with Interspersed Poisoned Data. arXiv preprint arXiv:2603.00516, 2026.

[arXiv preprint arXiv:2605.29286] Qian Wang, Zhongyi Tong, Nuo Chen, Zhaomin Wu, Bingsheng He. CrossAlpha: An Annual-Report Benchmark for Cross-Market Factor Research (with LLM Agents). arXiv preprint arXiv:2605.29286, 2026.

2025

[arXiv'25] Zhaomin Wu, Zhen Qin, Junyi Hou, Haodong Zhao, Qinbin Li, Bingsheng He, Lixin Fan. Vertical Federated Learning in Practice: The Good, the Bad, and the Ugly. arXiv preprint arXiv:2502.08160, 2025.

[arXiv'25] Haodong Zhao, Jidong Li, Zhaomin Wu†, Tianjie Ju, Zhuosheng Zhang, Bingsheng He, Gongshen Liu†. Disagreements in Reasoning: How a Model's Thinking Process Dictates Persuasion in Multi-Agent Systems. arXiv preprint arXiv:2509.21054, 2025.

[arXiv'25] Zhaomin Wu, Shida Wang, Ziyang Wang, Bingsheng He. Learning Relational Tabular Data without Shared Features. arXiv preprint arXiv:2502.10125, 2025.

EARLIER ACADEMIC HONORS

- **National Scholarship** (2017), Ministry of Education of China — top 1% at HUST.
- **Outstanding Student** (2016), HUST — top 1%.
- **Outstanding Undergraduate Thesis** (2019), HUST — top 5%.
- **Meritorious Winner, Mathematical Contest in Modeling** (2018), COMAP — top 10% worldwide.
- **Merit Student** (2016), HUST — top 5%.

PROFESSIONAL SERVICE & RECOGNITION

ICML 2026 Gold Reviewer Award: top 25% among reviewers.

Reviewer for conferences and journals:

2026: *Conf.* ICML, NeurIPS. *Journals* ACM Computing Surveys, TPDS, TKDE, TMC.
2025: *Conf.* ICLR, ICML, NeurIPS, AAAI, EMNLP. *Journals* TDSC, TPDS, TKDE, TAI.
2024: *Conf.* ICLR, NeurIPS, KDD, AISTATS. *Journals* TKDD, JPDC, TNNLS.
2023: *Conf.* NeurIPS, SIGMOD (artifact). *Journals* TPAMI, TKDE, TNNLS, IJCV, IoT-J.
2022: *Conf.* PAKDD. *Journals* TPDS, TIST.
2021: *Conf.* PAKDD.

Tutorial Speaker: IJCAI 2020.